Siyan(Evan) Wen

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Professional Summary:

Former Research Assistant in **Natural Language Processing** and team leader in a **Machine Learning competition** (3rd place among 9 teams).
Actively seeking full-time opportunities as a **Software Engineer**.
Familiar with Agile and Git development process. Proficient in **Microservices**, **kafka** message queue**, Maven** and **Gradle** build tools, **Junit Testing**, **Containerization**, Deployment and **CI/CD** pipeline.

Technical Skills:

Languages	java, C, C++, Python, Kotlin, Go, Ocaml, Assembly, JavaScript, TypeScript, PHP, Bash
Databases	MySQL, PostgreSQ, Cassandra, MongoDB, Redis
Backend	Spring Boot,,Spring Security, Kafka, Resilience4j, Django, Flask, Node.js, Express.js, Postman, Swagger
Frontend	React 18, TypeScript, HTML5, CSS3, AJAX
Mobile	Android Studio, Ktor, Jetpack Compose, Google Exoplayer, Room
Design Patterns	Singleton, Factory, Builder, Decorator, MVVM, MVC, CQRS, Saga
Cloud & DevOps	AWS (EC2, S3, RDS, EKS, ELB, IAM), GCP, Docker, Kubernetes, Helm, Jenkins, GitHub Actions, CI/CD
Testing	JUnit, Mockito, JaCoCo

Projects:

Food delivery platform

Technologies: ***Java, Spring Boot, Maven, Spring Security, Spring Cloud OpenFeign, Hibernate, Spring JPA, React.js, PostgreSQL, Cassandra, MongoDB, Junit, Mockito, Resilience4j, JaCoCo***

Frontend + backend: https://github.com/SiyanWen/shop-net demo video: https://tinyurl.com/yd982zm5

- Developed a full-stack e-commerce platform using **Spring Boot 3 microservices architecture**, comprising 5 domain services coordinated through a **Spring Cloud Gateway** API Gateway and **Netflix Eureka** service discovery.
- Applied **polyglot persistence** across services — PostgreSQL (JPA) for accounts and payments, MongoDB for product catalog, and Apache Cassandra for order storage — leveraging each database's strengths for its domain.
- Implemented **event-driven** order processing with **Apache Kafka** (KRaft mode): OrderService publishes order events and consumes payment confirmation events, decoupling the payment lifecycle via an asynchronous **Saga** pattern.
- Leveraged **Spring Cloud OpenFeign** for synchronous inter-service communication with **Resilience4j** circuit breakers and fallback factories, preventing cascade failures across service boundaries.
- Designed a **CQRS** pattern in OrderService using dual denormalized Cassandra tables (orders_by_id, orders_by_user) to optimize for distinct query access patterns without joins.
- Extracted a shared common-security library providing **JWT authentication** filters, HMAC-SHA token generation, and a **FeignAuthInterceptor** for service-to-service token propagation, enforcing consistent auth across all services.
- Built the frontend with **React 18** and **Ant Design**, proxied through the API Gateway, enabling cart management, order placement, and payment workflows.

Social media with AI image generation

Technologies: ***Go, React.js, DALL·E 3, GCP (App Engine, GCS, GCE), Elasticsearch***

Front End: *https://github.com/SiyanWen/AICanvas_front* *Back End:* https://github.com/SiyanWen/AICanvas_back

- Designed a social platform for generating and sharing AI-created images.
- Built a Go backend with **JWT authentication** and deployed on **Google Cloud App Engine**.
- Integrated OpenAI's **DALL·E 3** for prompt-based image generation.
- Implemented **Elasticsearch** for keyword and user-based post search.
- Managed image/video uploads with **Google Cloud Storage** buckets.

Android Music App

Technologies: ***Kotlin, Ktor, Jetpack Compose, Hilt Dependency Injection, Google Exoplayer, Room***

Front End: *https://github.com/SiyanWen/Spotify_front* *Back End:* *https://github.com/SiyanWen/Soptify_backend*

- Created a full-featured music app using Android Studio, Jetpack libraries and MVVM architecture.
- Supported song playback, album browsing, and favorites collection.
- Implemented **MVVM** architecture on Home/Album/Collection UI using **Jetpack Compose**.
- Enabled seamless navigation via **Jetpack Navigation Components**.
- Used RESTful API to handle requests from json-server by leveraging **Retrofit**.
- Enabled **Google Exoplayer** to handle music playback globally and Room for local caching.

Professional Experience:

Machine Learning competition Group Leader of 4 Apr. 2025-May. 2025

Stevens Institute of Technology. New Jersey, US

- After fully understanding the competition requirements, I took the initiative to assume the position of team leader, responsible for repeatedly confirming the details of the requirements with the competition initiator, setting up the environment, and reporting the results.
- Achieved **3rd place** among 9 teams based on test accuracy and feature reduction amount.
- Divided training data into 8 groups by performing **K-means clustering**.
- Owned 2 most challenging data groups (classification sets are heavily unbalanced). Reduced features by 98%. The train accuracy is 100% with stacked models.

NLP Research Assistant Jan. 2025-May. 2025

Stevens Institute of Technology. New Jersey, US

- Achieved significant speedup on annotation by developing a **Flask** python application to replace the commits (code block and commit messages) displayed in Excel table.
- The front end consists of commit code block, file hierarchy and timer. The annotator can set a time limit before starting, and each page will automatically turn after the time limit is reached. If no selection is made, the result of the current page will be set to null.
- Leveraged OpenAI API to perform **NER** (Named Entity Recognition) on text to generate training data. Displayed NER result in Excel by highlighting terms using Google Apps Script.
- Developed a **Beautiful Soup web crawler** to backup resources on website.

E-commerce Operations Specialist Oct. 2019-Mar. 2021

Shenzhen Liqian Technology Co. Ltd. Guangdong, China

- Identified high-potential products and optimized supply chains for Amazon storefronts.
- Conducted data-driven marketing and created promotional strategies to boost sales.
- Improved product page clarity with detailed specs and model guides, cutting return rates by 80%.

Education:

Stevens Institute of Technology *Hoboken, NJ*

Master of Science in Computer Science Sep.2023 - May.2025